AV FQS Model

Dynamic Out-of-Sample Validation Report

5-Year Rolling Period | Dynamically-Selected Prediction Dates | Nippon India Small Cap Fund & Mirae Asset Large & Mid Cap Fund

No date in this report is hardcoded. Each fund's prediction date is the latest cutoff at which it met the production eligibility rule (history ≥ rolling period + 2 years) AND had a complete 5-year future available in the dataset — both computed dynamically from the fund's own NAV history, per fund, using the production model's own rules and functions, never reimplemented.

Objective

This report does not compare fund RETURNS. It tests whether the AV FQS parameters computed from a fund's historical NAV data (Median WM, Blended WM, CV, Success %, FQS, category benchmark, and rank) successfully predict the SAME parameters, re-computed by the same production model, once the next 5-Year Rolling Period has actually happened. The core question is: does the historical AV FQS characterisation of a fund persist into its future characterisation?

Dynamic Date Selection — how each fund's dates were derived

Fund 1 — Nippon India Small Cap Fund: inception 2010-09-21 (detected from raw NAV history) → prediction date 2021-07-31 (latest cutoff satisfying both the 7-year minimum history rule and a complete 5-year future) → validation period 2021-08-01 to 2026-07-31.

Fund 2 — Mirae Asset Large & Mid Cap Fund: inception 2010-07-12 (detected from raw NAV history) → prediction date 2021-07-31 (latest cutoff satisfying both the 7-year minimum history rule and a complete 5-year future) → validation period 2021-08-01 to 2026-07-31.

Reusability

The function producing every number in this report, `backtest/report/dynamic_validation.py:validate_fund(scheme_code)`, takes only a scheme code. It contains no fund name, no date, and no benchmark value specific to either fund in this report. Running it against any other scheme code already present in the backtest's `nav_store` produces the same four tables with that fund's own dynamically-derived dates — extending to a fund not yet in the store requires running the existing, already-generic download and backtest pipeline for it, not new code.

Fund 1 — Nippon India Small Cap Fund

Nippon India Small Cap Fund - Growth Plan - Growth Option

Category: Small Cap (Equity Scheme - Small Cap Fund) — Scheme code 113177

Table 1 — Historical NAV Data & AV FQS Parameters

Historical Duration: 2010-09-21 → 2021-07-31 (fund's true inception, lineage-resolved, to the dynamically-selected prediction date)

Field	Value
Fund Name	Nippon India Small Cap Fund - Growth Plan - Growth Option
Category	Small Cap
Fund Inception Date	2010-09-21
Prediction Date (dynamic)	2021-07-31
Historical NAV Period	2010-09-21 to 2021-07-31
Total Historical NAV Count	2,673 daily NAVs (131 month-end points, 70 rolling 5Y windows)
Available History	10.83 years (minimum required: 7 years) — ELIGIBLE
WM (Median WM)	1.8621
Median WM	1.8621
Blended WM	1.5821
CV	0.2748
Success %	61.4%
Benchmark Type	Category minimum-WM hurdle (no benchmark FUND exists in production)
Benchmark WM	1.5600 (source: category_map.CATEGORY_TABLE)
Benchmark Median WM	N/A — see note below
Benchmark Blended WM	N/A — see note below
Benchmark CV	N/A — see note below
Benchmark Success %	N/A — see note below
FQS	59.3 (Weak)
Fund Rank	#2 of 164 eligible funds (by Blended WM)
Recommendation	Not recommended (rank #2 by Blended WM)

category_map.CATEGORY_TABLE defines a single numeric WM hurdle per category/horizon, not a benchmark fund — Median/Blended WM, CV, and Success % are properties of a FUND's window distribution and have no equivalent for a constant threshold.

Table 2 — AV FQS Prediction for the Next 5-Year Rolling Period

Field	Value
Prediction Generated On	2021-07-31
Prediction Date	2021-07-31
NAV Data Used to Generate This Prediction	2010-09-21 to 2021-07-31 (2,673 daily NAVs, 131 month-end points, 70 rolling 5Y windows) — identical to Table 1. ZERO NAVs from the predicted rolling period below were used.
Predicted Rolling Period (what the forecast is ABOUT, not what it was computed from)	2021-08-01 to 2026-07-31
Predicted WM	1.8621
Predicted Median WM	1.8621
Predicted Blended WM	1.5821
Predicted CV	0.2748
Predicted Success %	61.4%
Predicted Benchmark Type	Category minimum-WM hurdle (no benchmark FUND exists in production)
Predicted Benchmark WM	1.5600
Predicted Benchmark Median WM	N/A — see note, Table 1
Predicted Benchmark Blended WM	N/A — see note, Table 1
Predicted Benchmark CV	N/A — see note, Table 1
Predicted Benchmark Success %	N/A — see note, Table 1
Predicted FQS	59.3 (Weak)
Predicted Fund Rank	#2 of 164 eligible funds
Predicted Recommendation	Not recommended (rank #2 by Blended WM)

Generated using ONLY NAV data up to the prediction date — no NAV, rolling window, benchmark, ranking, or performance information from after this date was accessible at generation time.

Table 3 — Actual NAV Data & AV FQS Parameters

Field	Value
Actual Validation Duration	2021-08-01 to 2026-07-31
Total Validation NAV Count (NAVs strictly inside the validation window above)	1,231 daily NAVs
Actual WM (single realised 5Y SIP outcome)	1.4998 — computed using ONLY the 1,231 NAVs in the validation window above (the single SIP an investor starting the month after the prediction date would have completed).
NAV Data Used for Median/Blended WM, CV, Success %, FQS Below	2010-09-21 to 2026-07-31 (3,904 daily NAVs — the fund's FULL accumulated history, not just the validation window. score_horizon() always scores a fund's complete history; there is no production mode that scores an isolated slice, so this is the closest production-faithful "actual" for a distribution statistic.
Actual Median WM (self-persistence*)	1.9447
Actual Blended WM (self-persistence*)	1.6487
Actual CV (self-persistence*)	0.2384
Actual Success % (self-persistence*)	73.9%
Actual Benchmark Type	Category minimum-WM hurdle (no benchmark FUND exists in production)
Actual Benchmark WM	1.5600
Actual Benchmark Median WM	N/A — see note, Table 1
Actual Benchmark Blended WM	N/A — see note, Table 1
Actual Benchmark CV	N/A — see note, Table 1
Actual Benchmark Success %	N/A — see note, Table 1
Actual FQS (self-persistence*)	67.2 (Average)
Actual Fund Rank	#12 of 150 funds with a validated outcome at this cutoff
Actual Recommendation Outcome	Actual rank #12 of 150 validated

* Median WM / Blended WM / CV / Success % / FQS describe the SHAPE of a distribution of many rolling windows, not a single outcome — there is no valid "actual" value for a distribution statistic from one realised SIP. These rows are the SAME production score_horizon() calculation, re-run using the fund's complete accumulated history once prediction_date + 5 years of real data genuinely existed (exactly what the production app would show a user checking on that later date). "Actual WM" above the self-persistence block is the literal single realised outcome (backtest/actual_outcomes.py), included separately because it is the figure most directly comparable to what an investor actually experienced.

Note on Fund Rank denominator: 164 funds were eligible at the prediction date; only 150 of those ALSO have a complete, validated 5-year outcome as of the dataset's latest NAV date — the actual rank is computed among that smaller, fully-realised set. The two ranks are not on an identical denominator and Table 4 reports both denominators explicitly rather than treating the ranks as directly subtractable.

Table 4 — Prediction vs Actual Validation

Validation Period: 2021-08-01 to 2026-07-31

Parameter	Predicted	Actual	Abs. Diff	% Diff	Match
WM (single realised outcome)	1.8621	1.4998	-0.3623	-19.5%	NO MATCH
Median WM (self-persistence)	1.8621	1.9447	0.0826	4.4%	NO MATCH
Blended WM (self-persistence)	1.5821	1.6487	0.0666	4.2%	NO MATCH
CV (self-persistence)	0.2748	0.2384	-0.0364	-13.2%	NO MATCH
Success % (self-persistence)	61.4%	73.9%	12.4%	20.2%	NO MATCH
Benchmark WM	1.5600	1.5600	0.0000	0.0%	MATCH
Benchmark Median WM	N/A	N/A	N/A	N/A	N/A
Benchmark Blended WM	N/A	N/A	N/A	N/A	N/A
Benchmark CV	N/A	N/A	N/A	N/A	N/A
Benchmark Success %	N/A	N/A	N/A	N/A	N/A
FQS	59.3	67.2	7.9	13.3%	NO MATCH

Parameter	Predicted	Actual	Match
Fund Rank	#2 of 164	#12 of 150	Rank moved by +10 places
Recommendation	Not recommended (rank #2 by Blended WM)	Actual rank #12 of 150 validated	MATCH
Benchmark Cleared	Yes	Yes	MATCH

Fund 2 — Mirae Asset Large & Mid Cap Fund

Mirae Asset Large & Midcap Fund - Regular Plan - Growth

Category: Large Cap (Equity Scheme - Large & Mid Cap Fund) — Scheme code 112932

Table 1 — Historical NAV Data & AV FQS Parameters

Historical Duration: 2010-07-12 → 2021-07-31 (fund's true inception, lineage-resolved, to the dynamically-selected prediction date)

Field	Value
Fund Name	Mirae Asset Large & Midcap Fund - Regular Plan - Growth
Category	Large Cap
Fund Inception Date	2010-07-12
Prediction Date (dynamic)	2021-07-31
Historical NAV Period	2010-07-12 to 2021-07-31
Total Historical NAV Count	2,724 daily NAVs (133 month-end points, 72 rolling 5Y windows)
Available History	11.0 years (minimum required: 7 years) — ELIGIBLE
WM (Median WM)	1.7793
Median WM	1.7793
Blended WM	1.6020
CV	0.1982
Success %	86.1%
Benchmark Type	Category minimum-WM hurdle (no benchmark FUND exists in production)
Benchmark WM	1.3617 (source: category_map.CATEGORY_TABLE)
Benchmark Median WM	N/A — see note below
Benchmark Blended WM	N/A — see note below
Benchmark CV	N/A — see note below
Benchmark Success %	N/A — see note below
FQS	81.8 (Strong)
Fund Rank	#1 of 164 eligible funds (by Blended WM)
Recommendation	Recommended (rank #1 by Blended WM)

category_map.CATEGORY_TABLE defines a single numeric WM hurdle per category/horizon, not a benchmark fund — Median/Blended WM, CV, and Success % are properties of a FUND's window distribution and have no equivalent for a constant threshold.

Table 2 — AV FQS Prediction for the Next 5-Year Rolling Period

Field	Value
Prediction Generated On	2021-07-31
Prediction Date	2021-07-31
NAV Data Used to Generate This Prediction	2010-07-12 to 2021-07-31 (2,724 daily NAVs, 133 month-end points, 72 rolling 5Y windows) — identical to Table 1. ZERO NAVs from the predicted rolling period below were used.
Predicted Rolling Period (what the forecast is ABOUT, not what it was computed from)	2021-08-01 to 2026-07-31
Predicted WM	1.7793
Predicted Median WM	1.7793
Predicted Blended WM	1.6020
Predicted CV	0.1982
Predicted Success %	86.1%
Predicted Benchmark Type	Category minimum-WM hurdle (no benchmark FUND exists in production)
Predicted Benchmark WM	1.3617
Predicted Benchmark Median WM	N/A — see note, Table 1
Predicted Benchmark Blended WM	N/A — see note, Table 1
Predicted Benchmark CV	N/A — see note, Table 1
Predicted Benchmark Success %	N/A — see note, Table 1
Predicted FQS	81.8 (Strong)
Predicted Fund Rank	#1 of 164 eligible funds
Predicted Recommendation	Recommended (rank #1 by Blended WM)

Generated using ONLY NAV data up to the prediction date — no NAV, rolling window, benchmark, ranking, or performance information from after this date was accessible at generation time.

Table 3 — Actual NAV Data & AV FQS Parameters

Field	Value
Actual Validation Duration	2021-08-01 to 2026-07-31
Total Validation NAV Count (NAVs strictly inside the validation window above)	1,231 daily NAVs
Actual WM (single realised 5Y SIP outcome)	1.3389 — computed using ONLY the 1,231 NAVs in the validation window above (the single SIP an investor starting the month after the prediction date would have completed).
NAV Data Used for Median/Blended WM, CV, Success %, FQS Below	2010-07-12 to 2026-07-31 (3,955 daily NAVs — the fund's FULL accumulated history, not just the validation window. score_horizon() always scores a fund's complete history; there is no production mode that scores an isolated slice, so this is the closest production-faithful "actual" for a distribution statistic.
Actual Median WM (self-persistence*)	1.6301
Actual Blended WM (self-persistence*)	1.5147
Actual CV (self-persistence*)	0.1773
Actual Success % (self-persistence*)	88.6%
Actual Benchmark Type	Category minimum-WM hurdle (no benchmark FUND exists in production)
Actual Benchmark WM	1.3617
Actual Benchmark Median WM	N/A — see note, Table 1
Actual Benchmark Blended WM	N/A — see note, Table 1
Actual Benchmark CV	N/A — see note, Table 1
Actual Benchmark Success %	N/A — see note, Table 1
Actual FQS (self-persistence*)	82.4 (Strong)
Actual Fund Rank	#62 of 150 funds with a validated outcome at this cutoff
Actual Recommendation Outcome	Actual rank #62 of 150 validated

* Median WM / Blended WM / CV / Success % / FQS describe the SHAPE of a distribution of many rolling windows, not a single outcome — there is no valid "actual" value for a distribution statistic from one realised SIP. These rows are the SAME production score_horizon() calculation, re-run using the fund's complete accumulated history once prediction_date + 5 years of real data genuinely existed (exactly what the production app would show a user checking on that later date). "Actual WM" above the self-persistence block is the literal single realised outcome (backtest/actual_outcomes.py), included separately because it is the figure most directly comparable to what an investor actually experienced.

Note on Fund Rank denominator: 164 funds were eligible at the prediction date; only 150 of those ALSO have a complete, validated 5-year outcome as of the dataset's latest NAV date — the actual rank is computed among that smaller, fully-realised set. The two ranks are not on an identical denominator and Table 4 reports both denominators explicitly rather than treating the ranks as directly subtractable.

Table 4 — Prediction vs Actual Validation

Validation Period: 2021-08-01 to 2026-07-31

Parameter	Predicted	Actual	Abs. Diff	% Diff	Match
WM (single realised outcome)	1.7793	1.3389	-0.4404	-24.8%	NO MATCH
Median WM (self-persistence)	1.7793	1.6301	-0.1492	-8.4%	NO MATCH
Blended WM (self-persistence)	1.6020	1.5147	-0.0873	-5.4%	NO MATCH
CV (self-persistence)	0.1982	0.1773	-0.0209	-10.5%	NO MATCH
Success % (self-persistence)	86.1%	88.6%	2.5%	2.9%	NO MATCH
Benchmark WM	1.3617	1.3617	0.0000	0.0%	MATCH
Benchmark Median WM	N/A	N/A	N/A	N/A	N/A
Benchmark Blended WM	N/A	N/A	N/A	N/A	N/A
Benchmark CV	N/A	N/A	N/A	N/A	N/A
Benchmark Success %	N/A	N/A	N/A	N/A	N/A
FQS	81.8	82.4	0.6	0.7%	NO MATCH

Parameter	Predicted	Actual	Match
Fund Rank	#1 of 164	#62 of 150	Rank moved by +61 places
Recommendation	Recommended (rank #1 by Blended WM)	Actual rank #62 of 150 validated	NO MATCH
Benchmark Cleared	Yes	Yes	MATCH

Validation Checklist

Item	Evidence
✓ No future NAV data used during prediction	Confirmed by construction: point_in_time.truncate() slices every fund's NAV to date <= cutoff before any calculation runs, verified by backtest_engine.py's ordering comment and cross-checked exactly (fast path vs full recomputation) before this run.
✓ Dynamic date selection implemented	prediction_date = max(cutoff_date) among each fund's own validated rows in backtest_comparison.parquet — no literal date in dynamic_validation.py.
✓ Prediction date calculated programmatically	See above; identical code path for both funds, producing 2021-07-31 for each because that is the true latest-complete-future boundary given the dataset's own NAV ceiling, not a coincidence of hardcoding.
✓ Historical period calculated programmatically	fund_inception_date read from each fund's own first NAV row (lineage-resolved); end date = the dynamic prediction date.
✓ Validation period calculated programmatically	prediction_date + 1 day → prediction_date + horizon_years, via pandas DateOffset.
✓ Eligibility rule applied dynamically	backtest/eligibility.py assess(), reusing server.py's own history_years >= horizon + HISTORY_BUFFER_YEARS rule, evaluated fresh at each fund's own cutoff.
✓ Production rolling calculations reused	backend/fqs_engine.py compute_rolling_windows() via backtest/point_in_time.py, unmodified.
✓ Production WM / Median WM / Blended WM reused	backend/fqs_engine.py score_horizon(), called via backtest/model_adapter.py predict().
✓ Production CV reused	score_horizon()'s own cv field, unmodified.
✓ Production Success % reused	score_horizon()'s own success_pct field, unmodified.
✓ Production Benchmark Selection reused	backend/category_map.py min_wm_for(), the exact function backtest/point_in_time.py calls — independently spot-checked against category_map.min_wm_for('Small Cap', 5) and ('Large Cap', 5) directly, matching the backtest output exactly.
✓ Production FQS reused	score_horizon()'s own fqs field, unmodified.
✓ Production Ranking reused	backtest/backtest_engine.py rank() [predicted, by blended_wm] and rank_actuals() [actual, by realised WM] — the same functions used throughout the 164-fund backtest.
✓ Production Recommendation reused	recommended_flag (rank #1 by Blended WM) — disclosed as the closest production-adjacent concept, since the app itself has no explicit Buy/Hold/Avoid object (MODEL_AUDIT.md §0).
✓ No hardcoded dates	Confirmed — see Dynamic Date Selection section above.
✓ No hardcoded benchmarks	Confirmed — benchmark_min_wm is looked up from category_map.CATEGORY_TABLE per fund's own category and horizon, never a literal number in this report's code.
✓ No recreated formulas	Every WM/CV/Success%/FQS/benchmark/rank value traces to an unmodified production function call, listed per-field in each fund's checklist_evidence block in dynamic_validation_data.json.
✓ All calculations match production implementation exactly	This run reuses the SAME backtest_predictions.parquet / backtest_comparison.parquet already cross-checked fund-by-fund (fast path vs full recomputation, 8/8 funds, exact match) before being trusted for the prior 164-fund report.

Full machine-readable output, including every field cited above with its exact production source, accompanies this report as dynamic_validation_data.json.